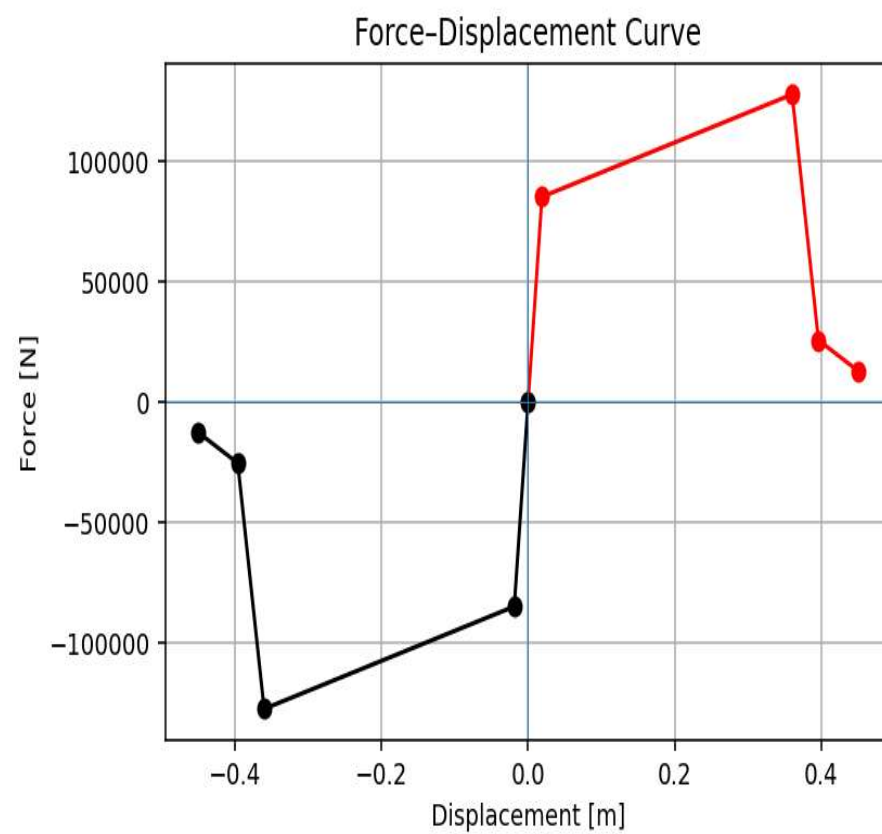
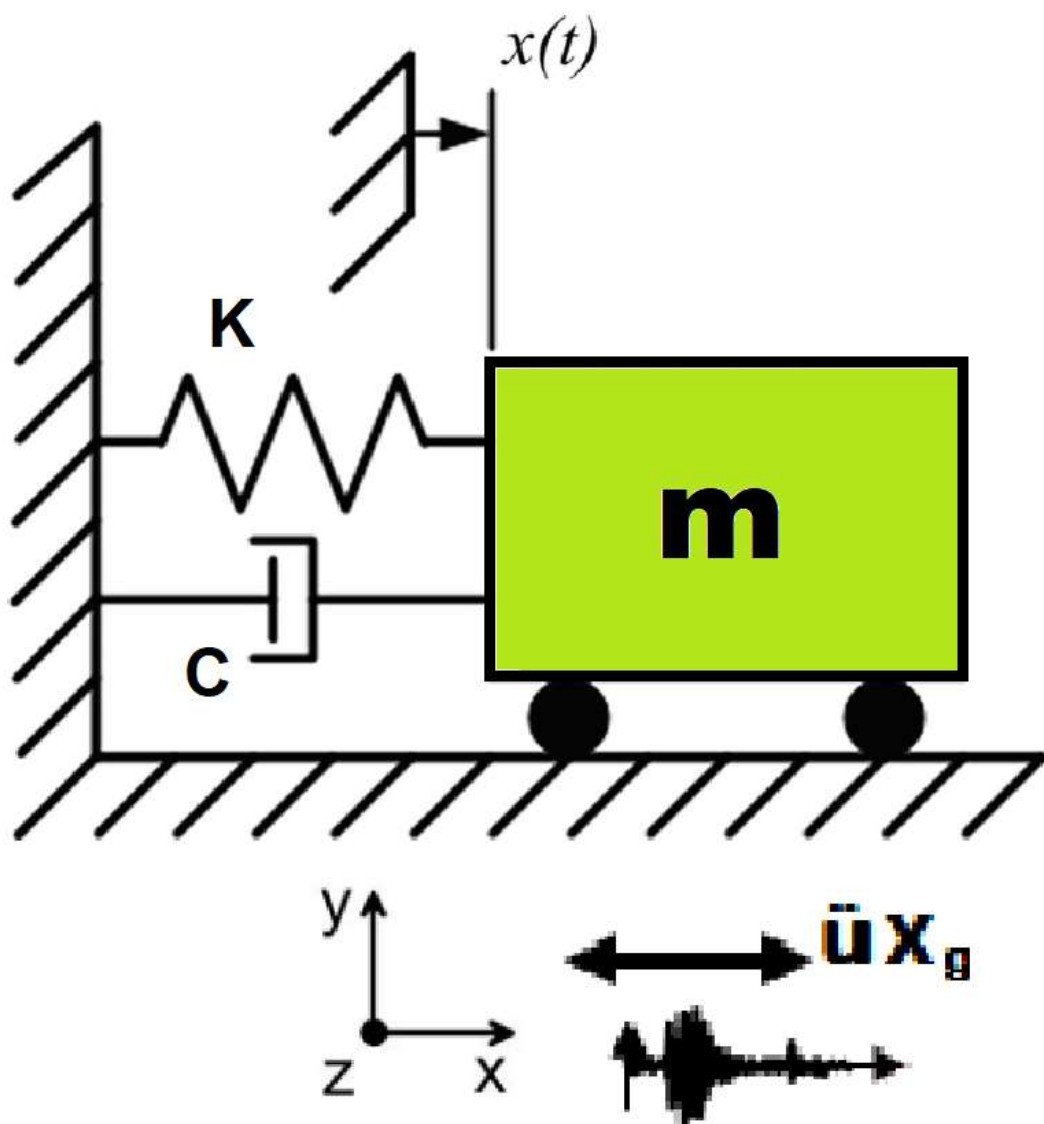


>> IN THE NAME OF ALLAH, THE MOST GRACIOUS, THE MOST MERCIFUL <<

**FRAGILITY ANALYSIS BASED ON
ACCELERATION, STRUCTURAL DUCTILITY
DAMAGE INDEX, ENERGY DISSIPATION
CAPACITY INDEX, EQUIVALENT VISCOUS
DAMPING RATIO WITH NONLINEAR DYNAMIC
ANALYSIS (IDA) OF A SINGLE-DEGREE-OF-
FREEDOM (SDOF) SYSTEM UTILIZING 30
GROUND MOTIONS IN OPENSEES**

WRITTEN BY SALAR DELAVAR GHASHGHAEI (QASHQAI)



$$\text{Structural Ductility Damage Index} = \frac{\Delta_d - \Delta_y}{\Delta_u - \Delta_y}$$

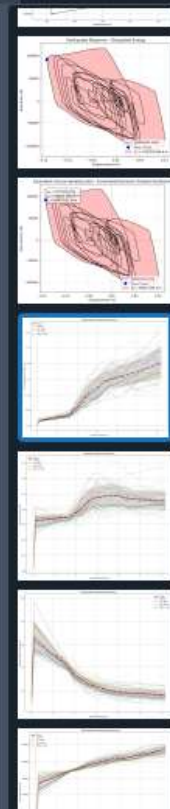
Δ_d = Lateral Displacement from Dynamic Analysis

Δ_y = Lateral Yield Displacement from Pushover Analysis

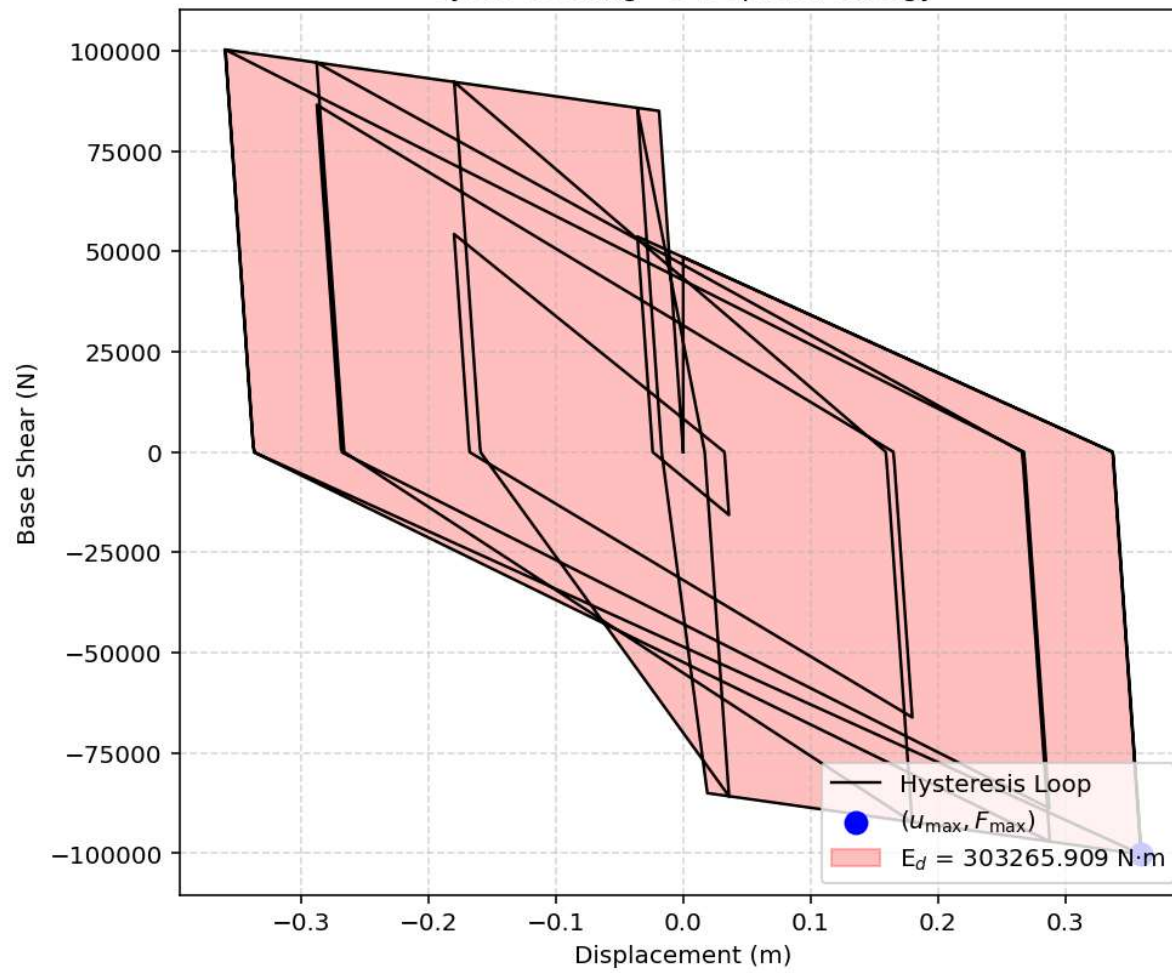
Δ_u = Lateral Ultimate Displacement from Pushover Analysis

SDOF SDOF RESPONSE SPECTRUM EDCI EVDR.py X EQUIVALENT VISCOUS DAMPING RATIO FUN.py X

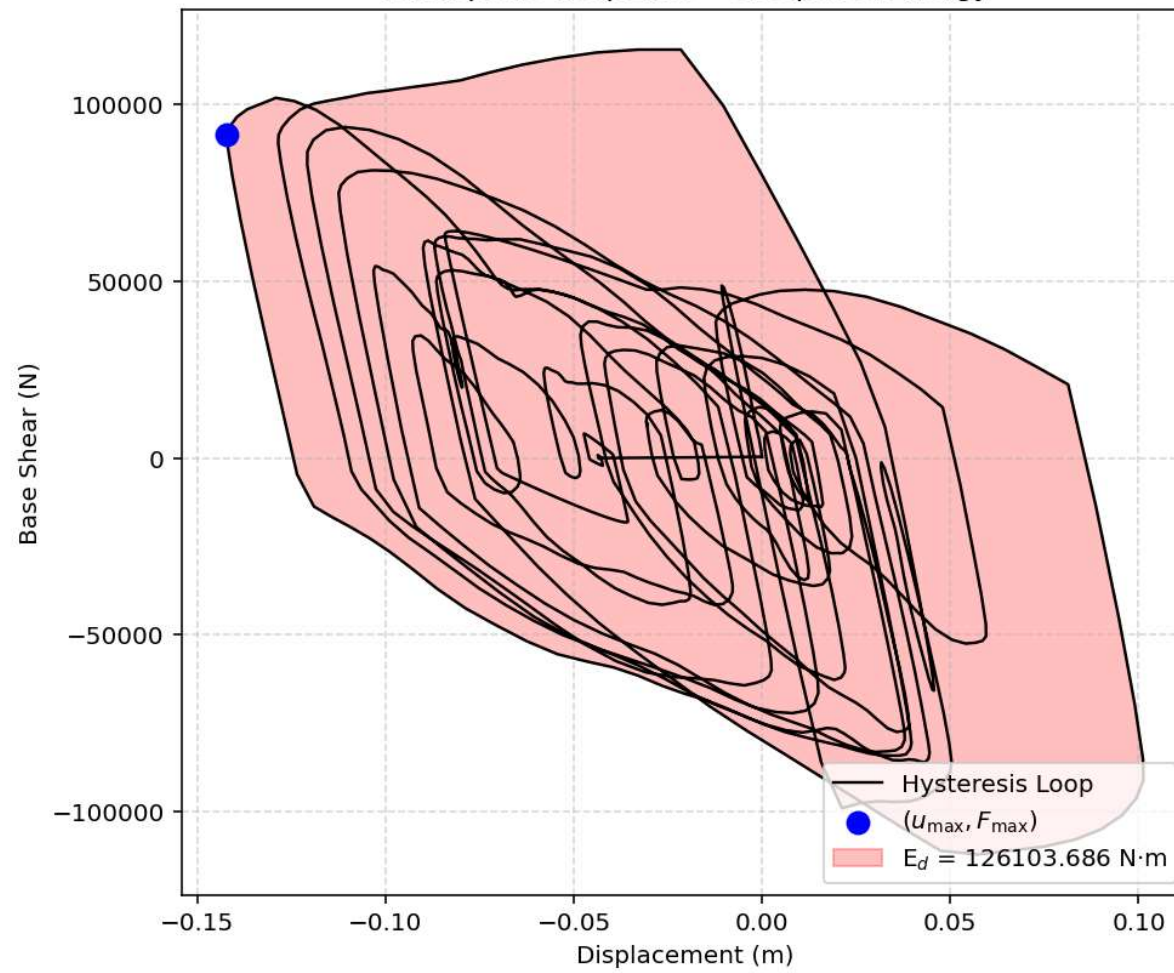
Options



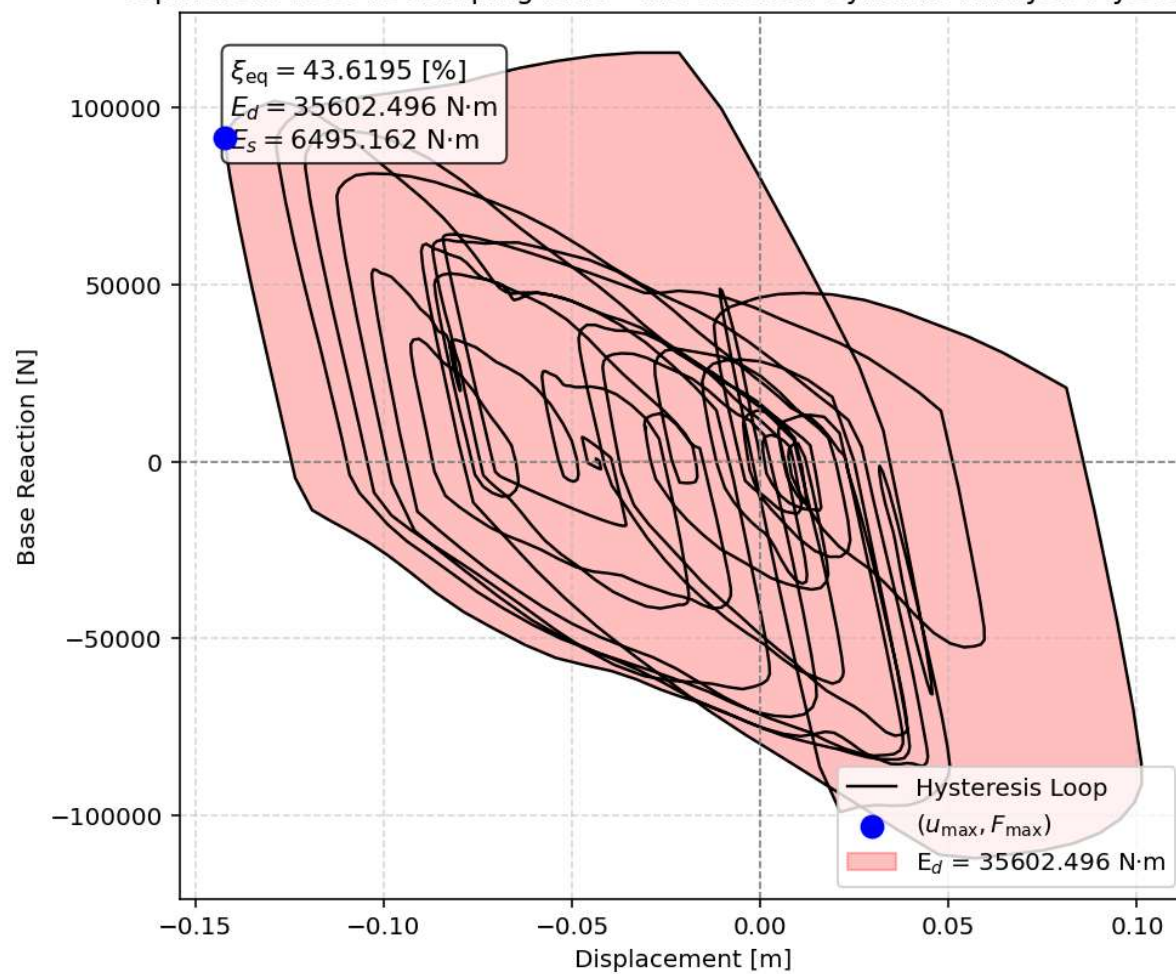
Cyclic Loading - Dissipated Energy

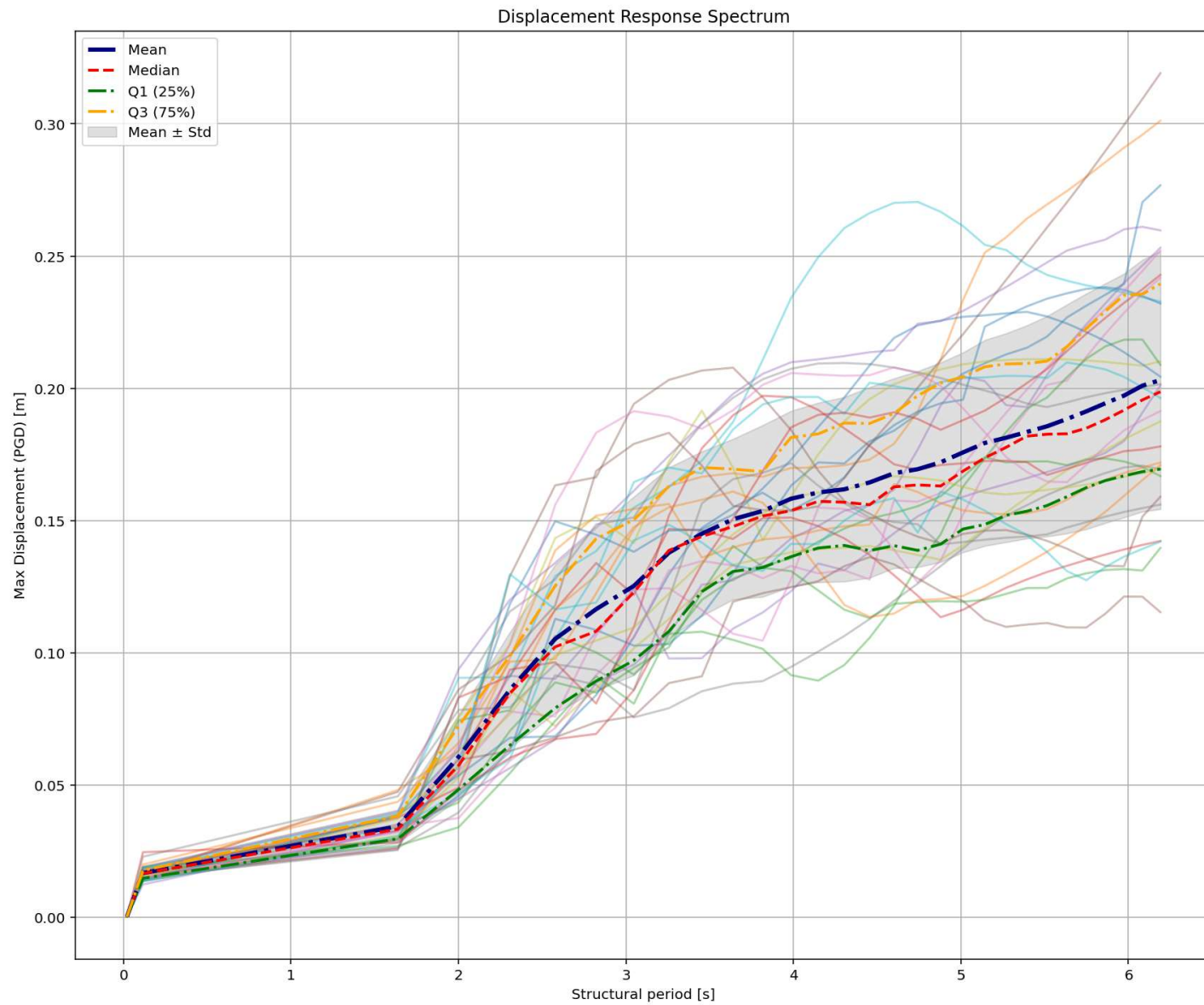


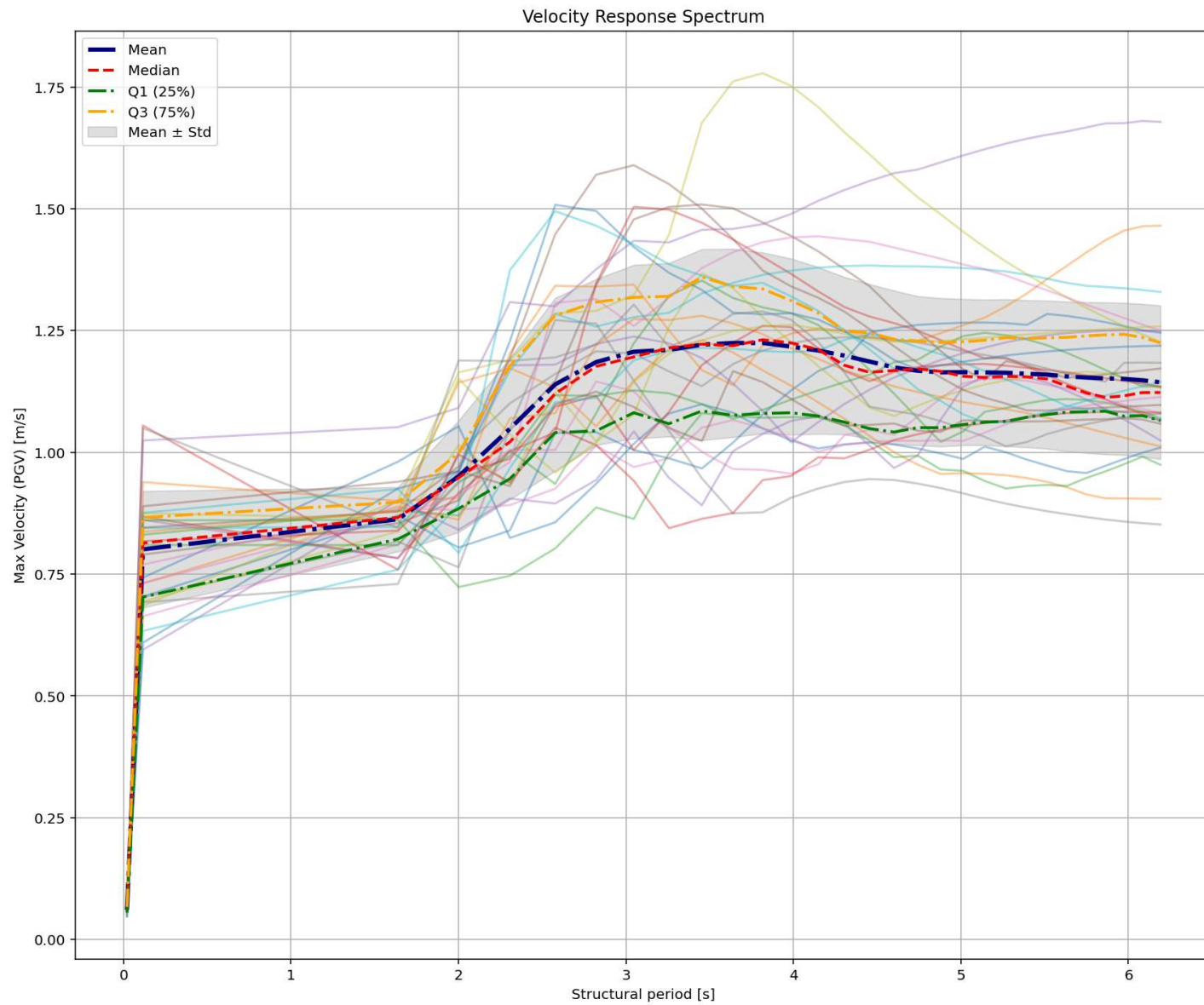
Earthquake Response - Dissipated Energy

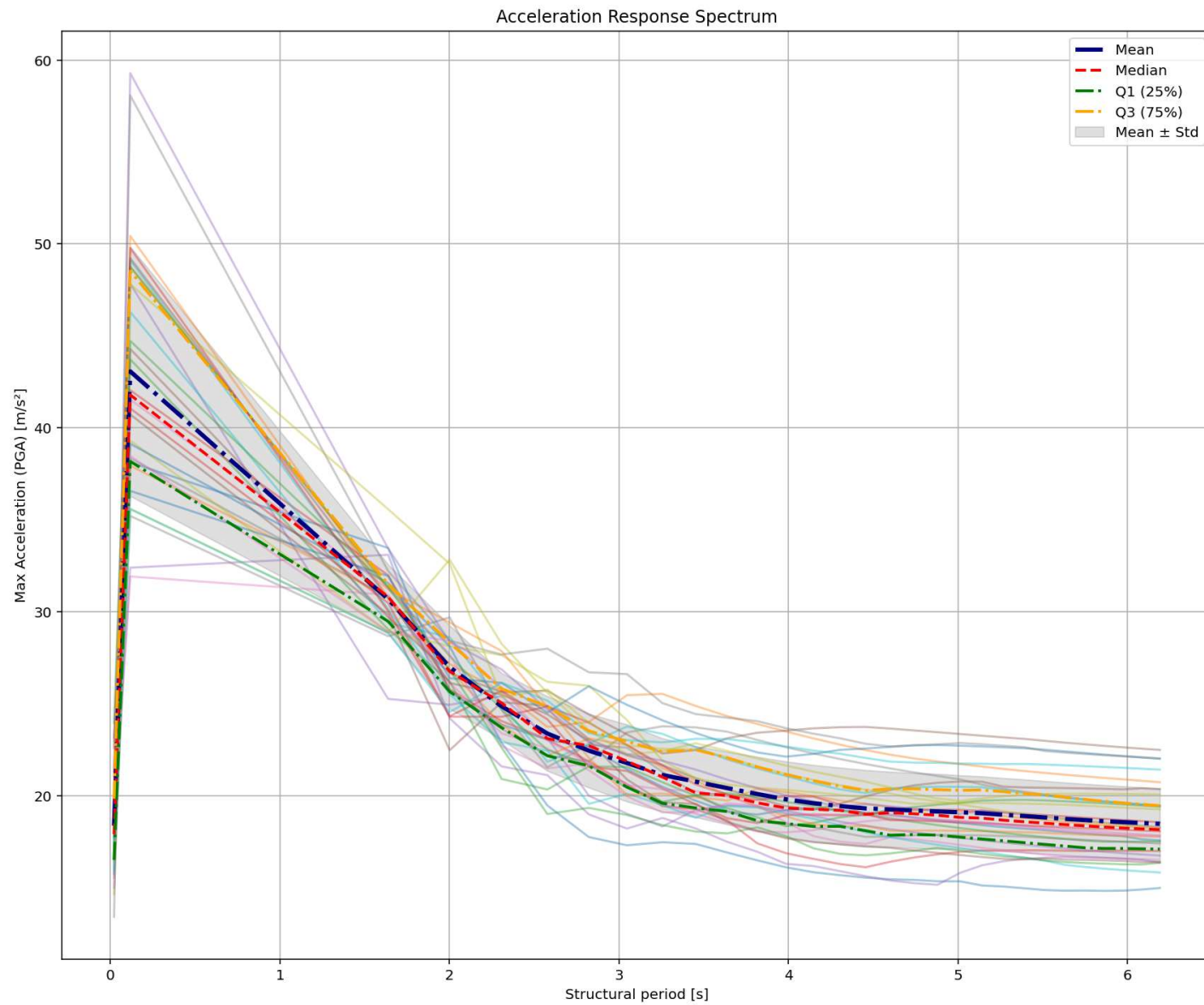


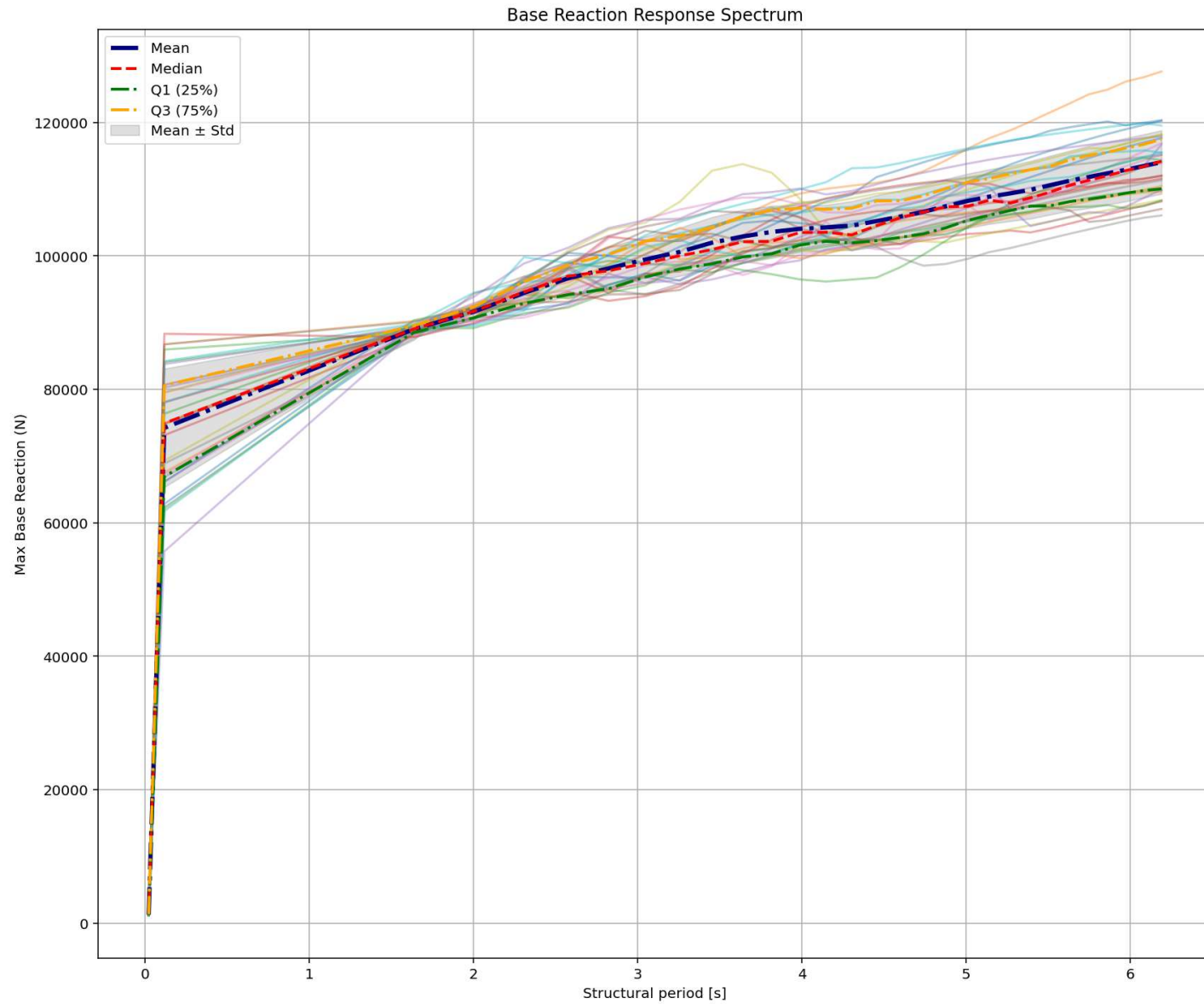
Equivalent viscous damping ratio - Incremental Dynamic Analysis Hysteresis

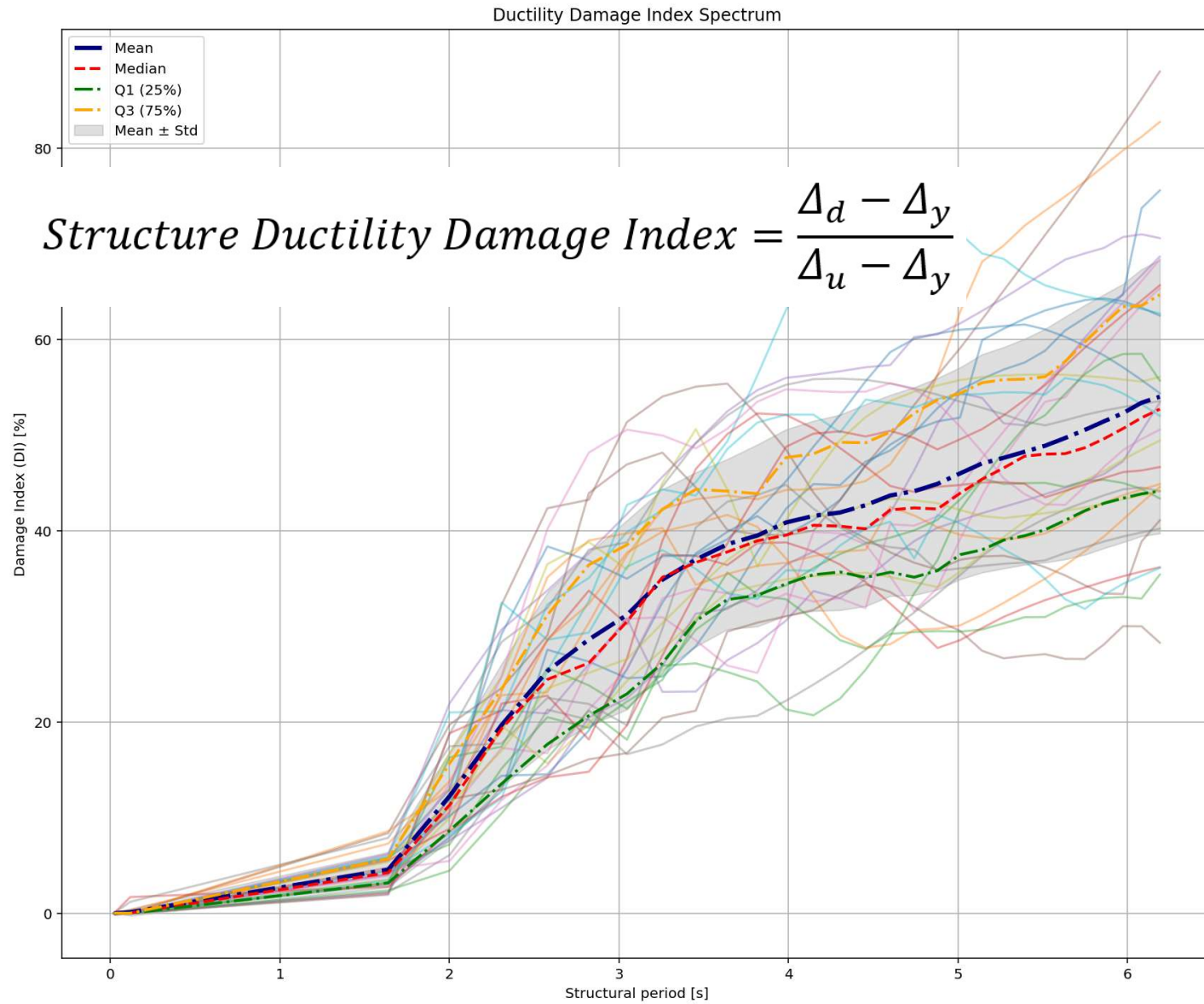


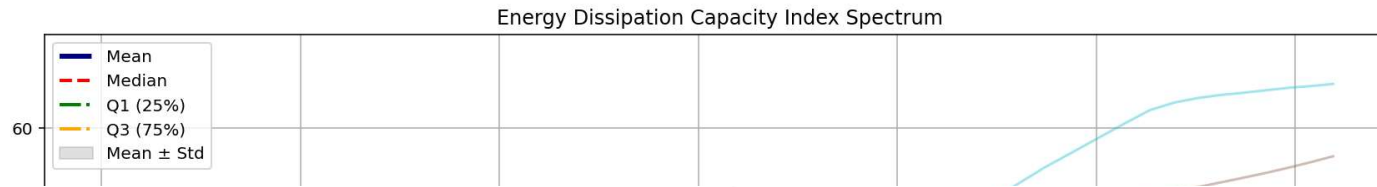




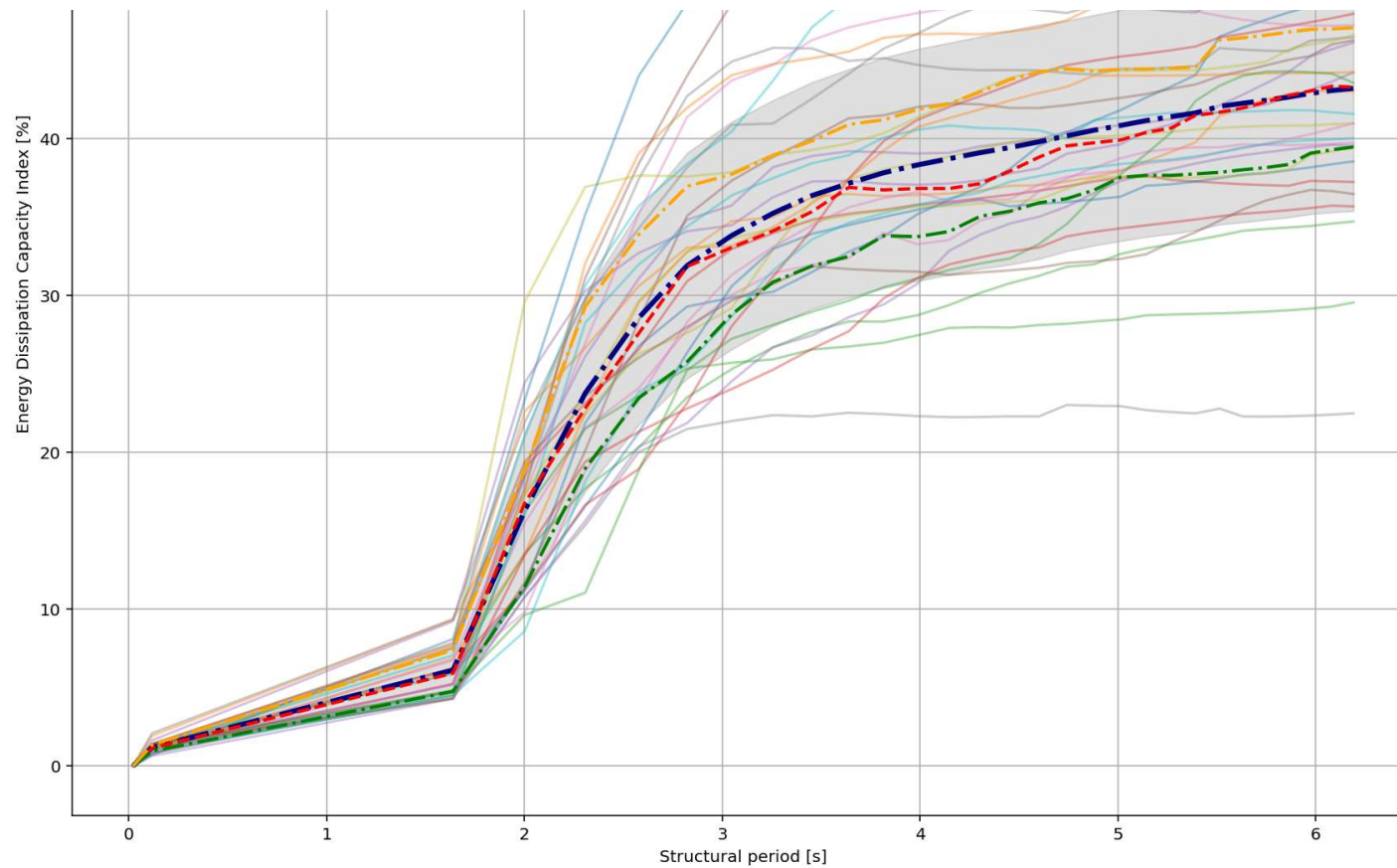


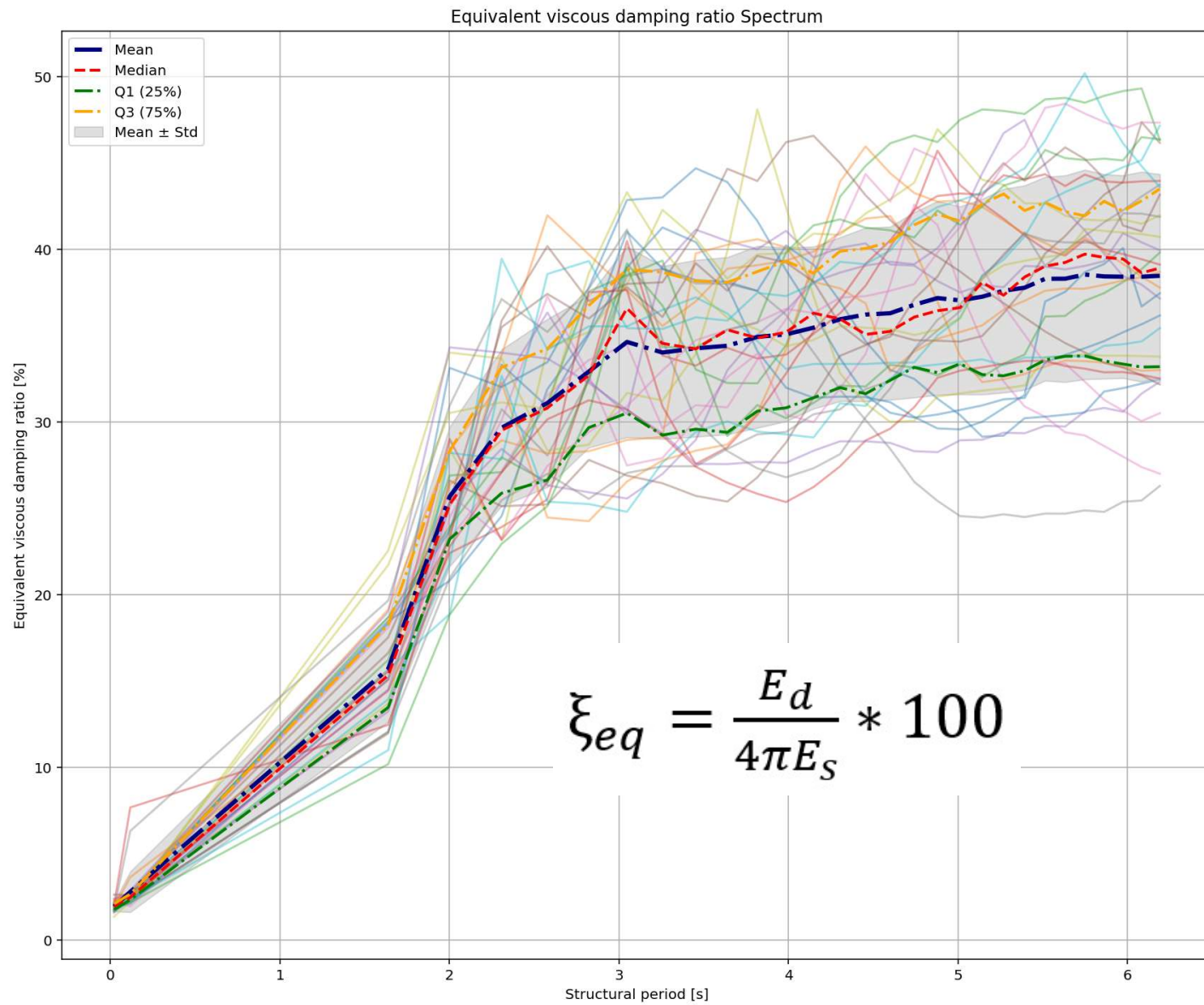




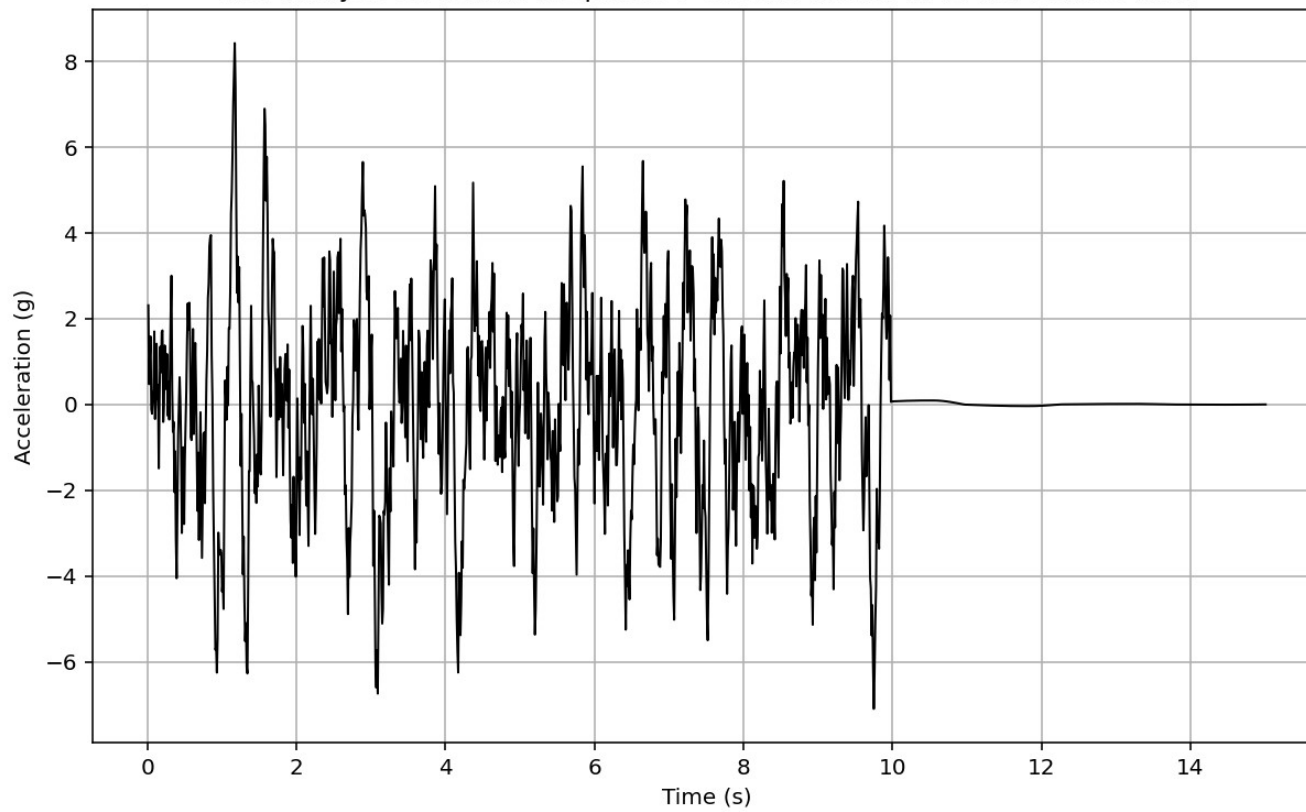


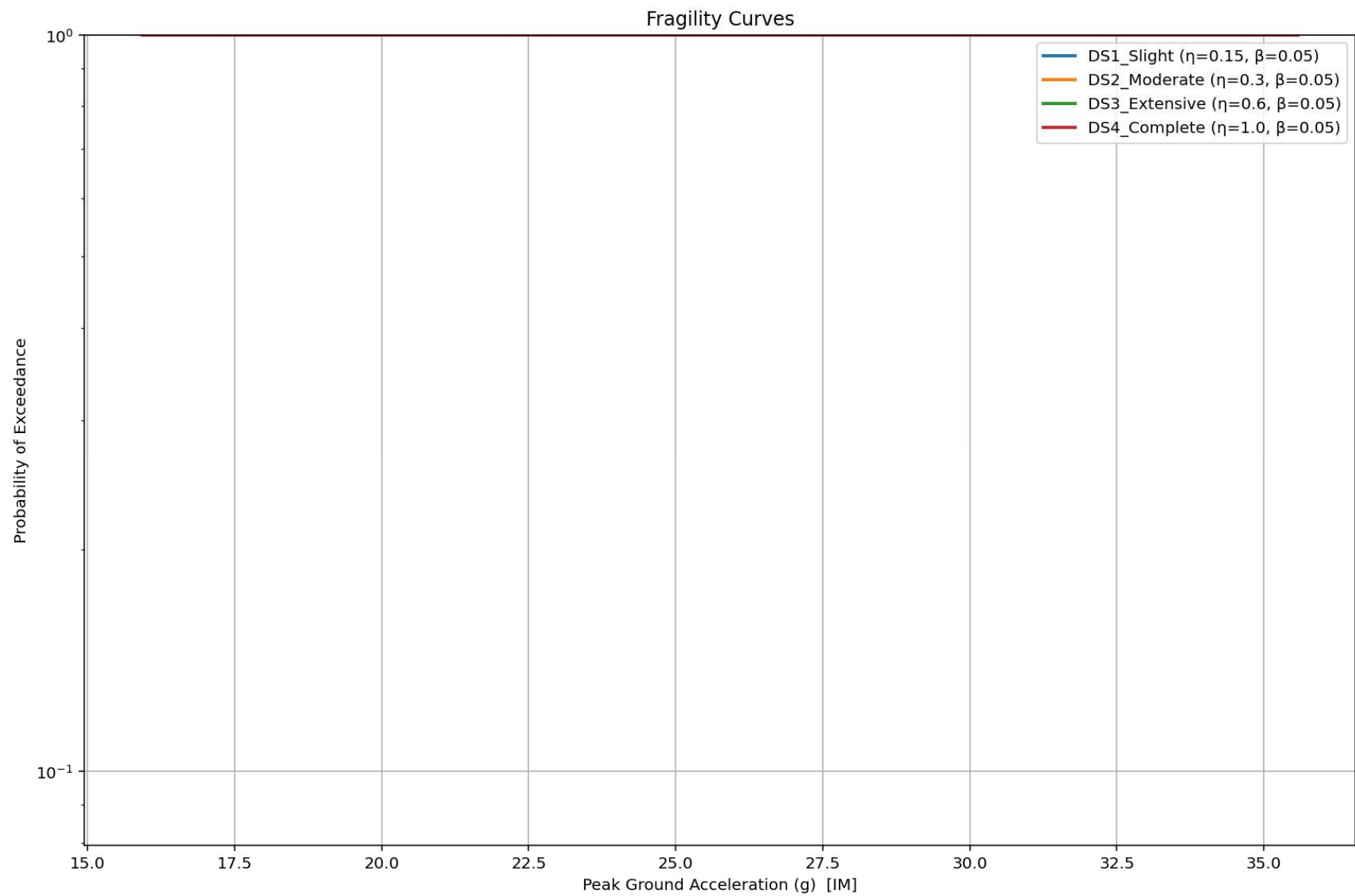
$$\text{Energy Dissipation Capacity Index} = \frac{E_d(\text{Dissipated Energy from Earthquake})}{E_d(\text{Dissipated Energy from Cyclic Displacement})} * 100$$



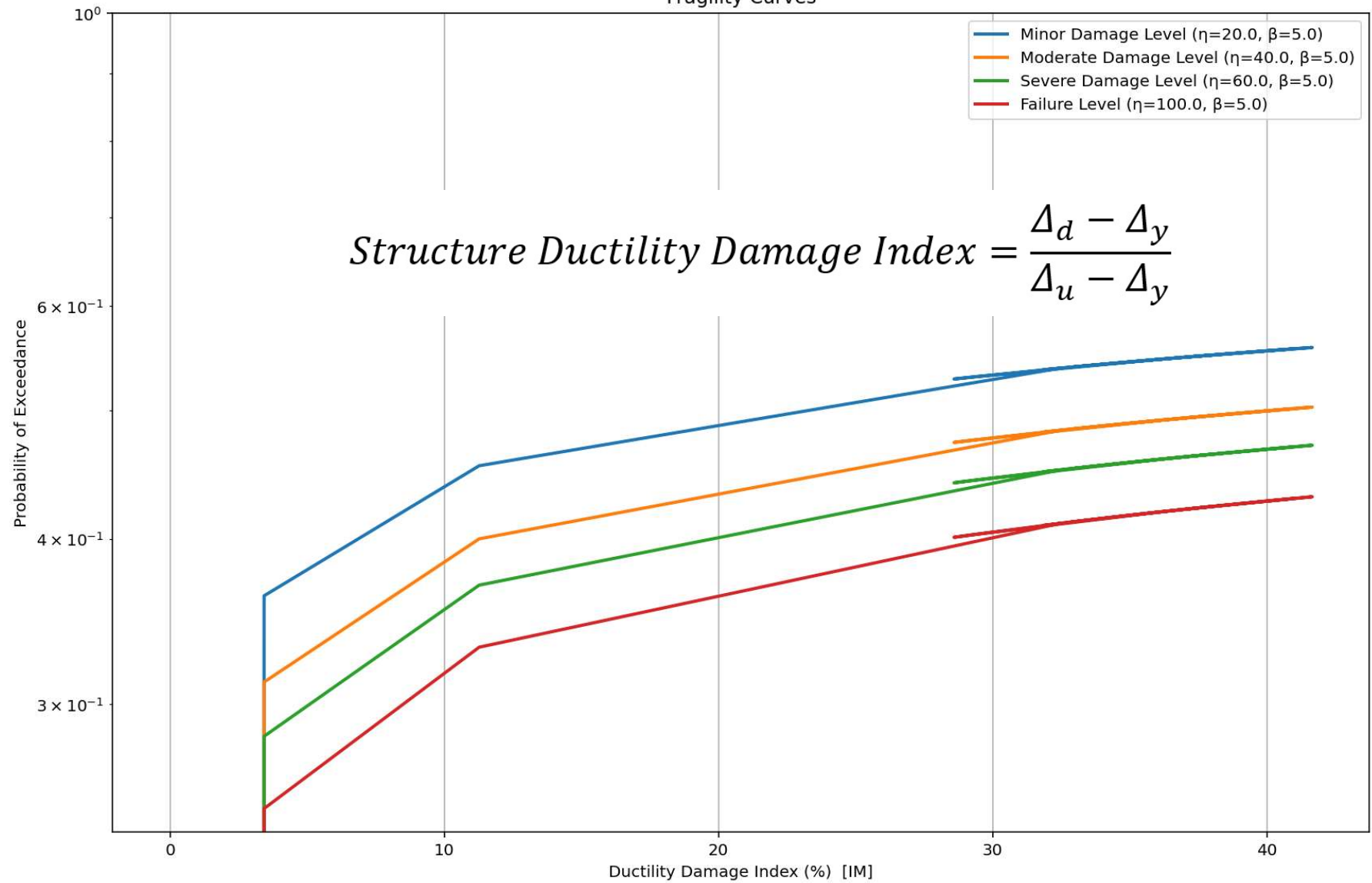


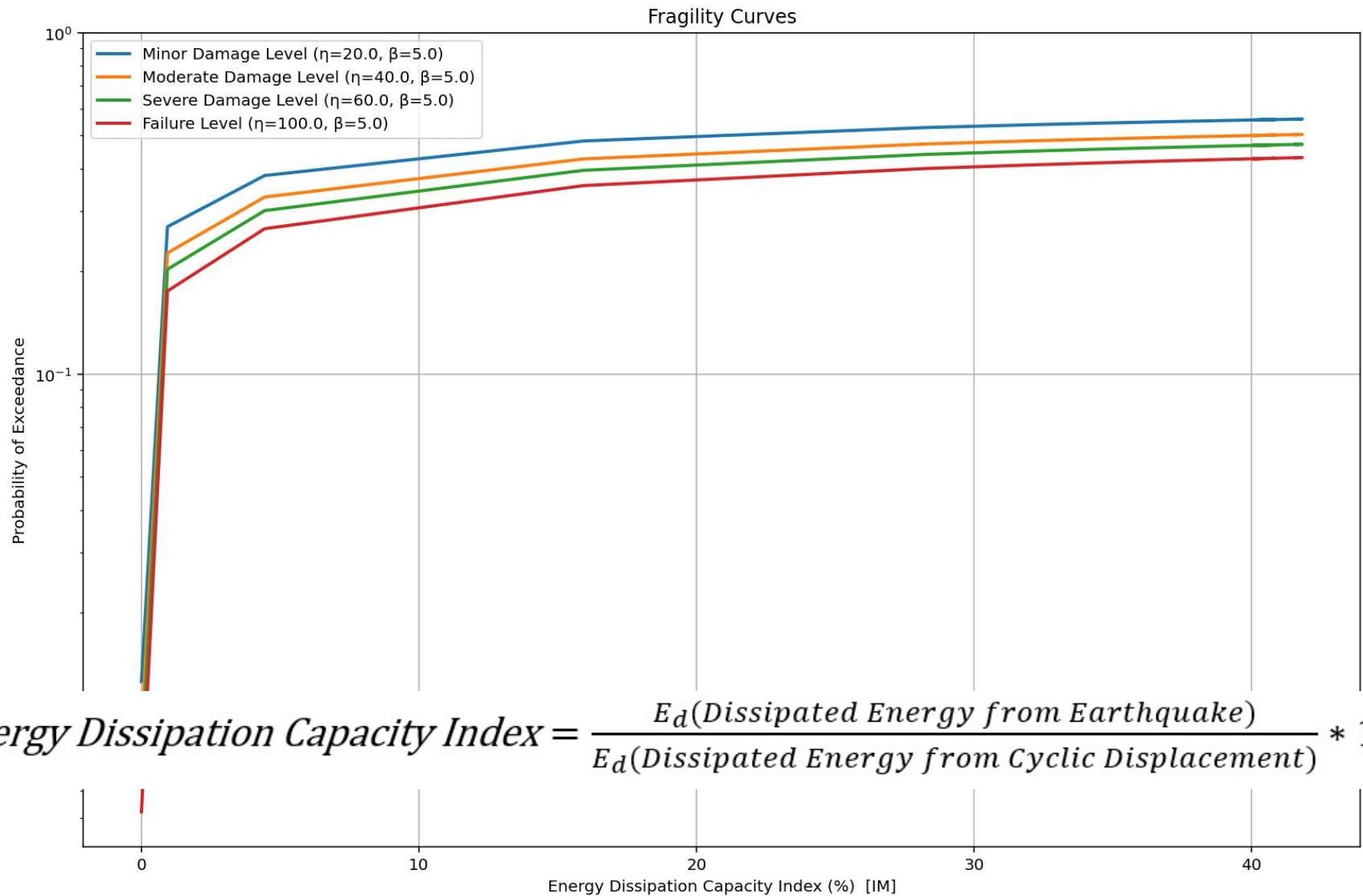
Last Analysis Structural Response + Ground Motion ::: MAX. ABS. : 8.4259



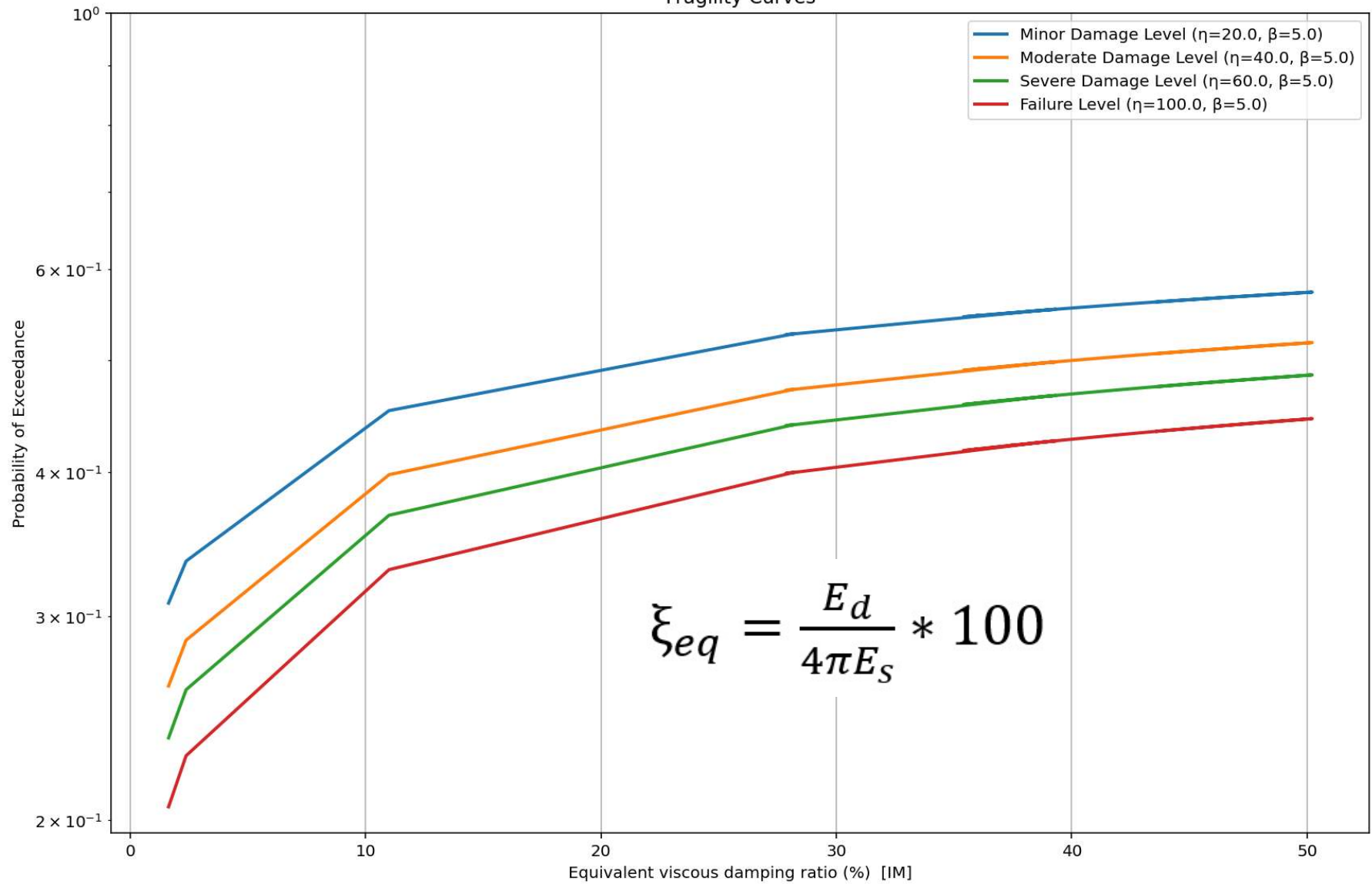


Fragility Curves

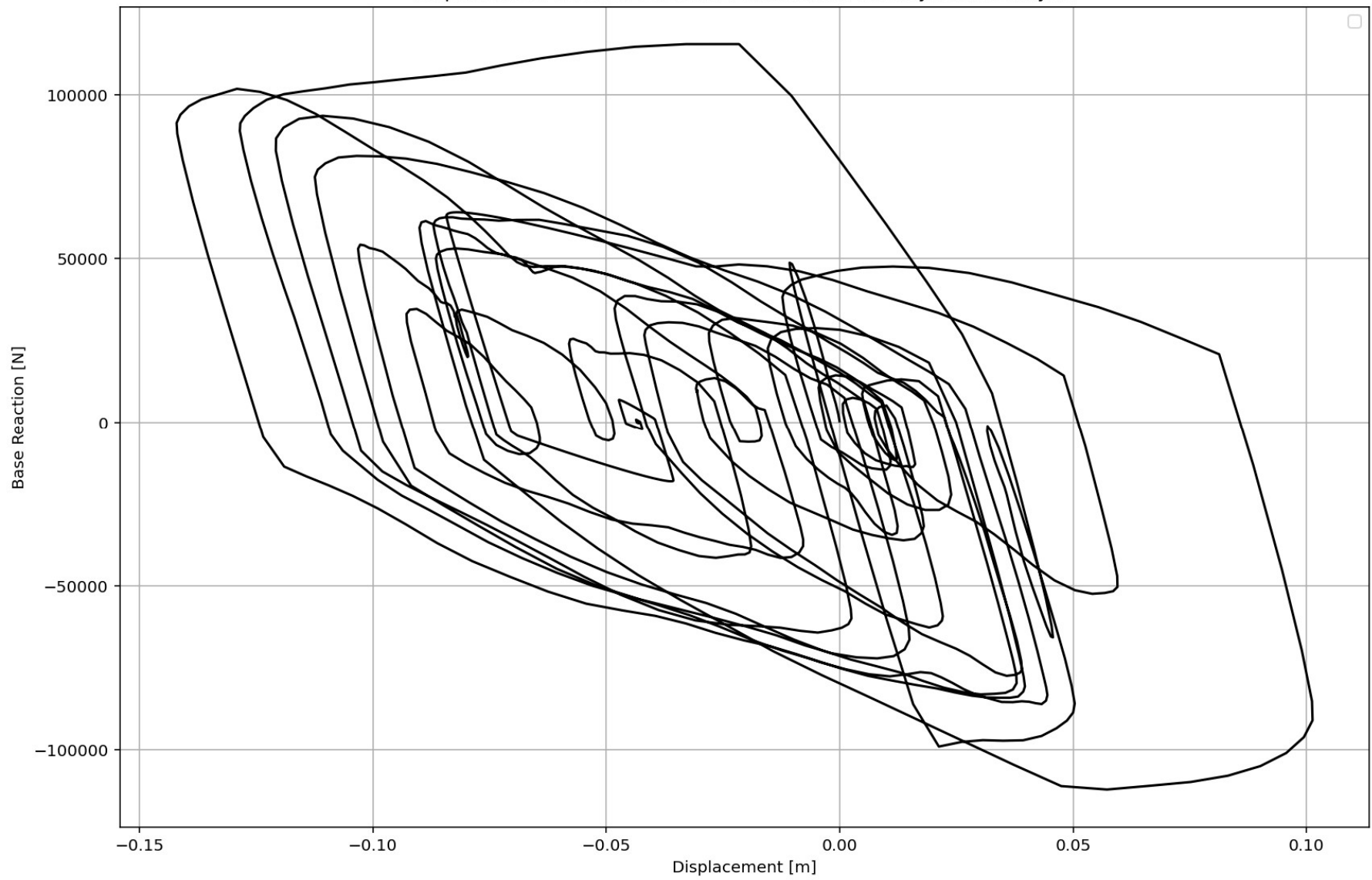


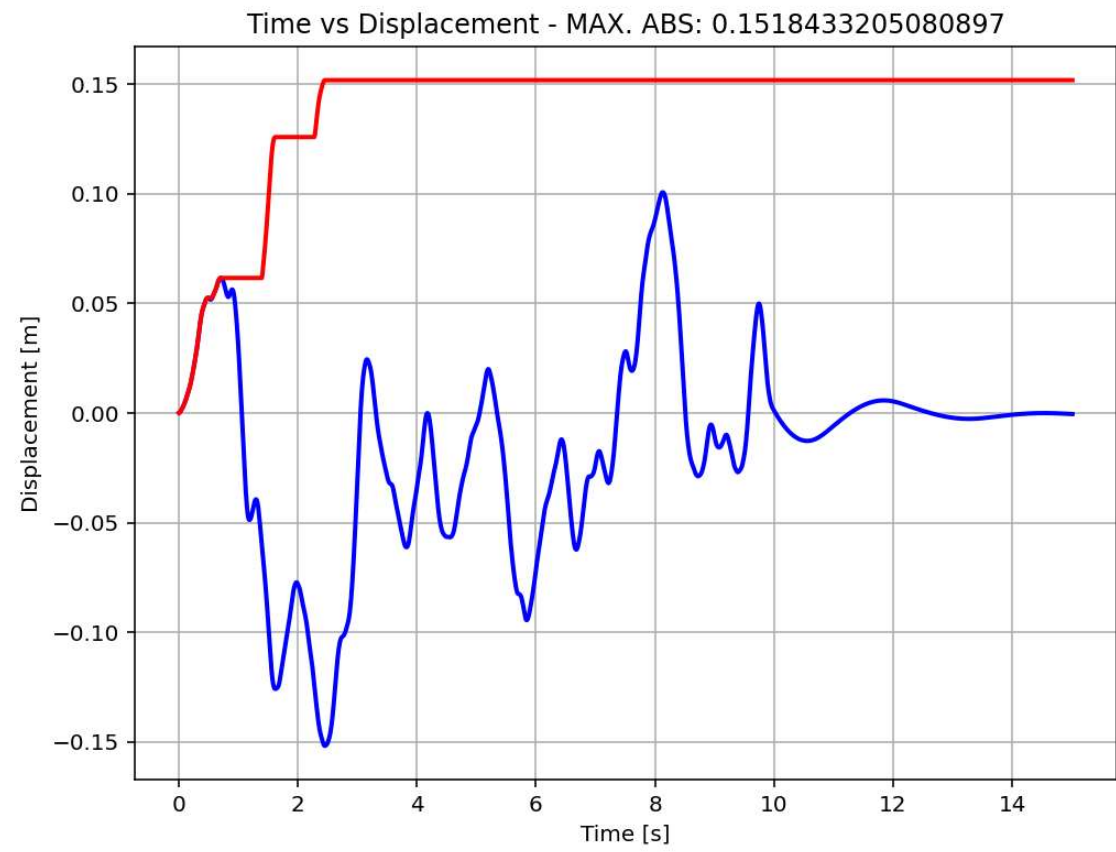


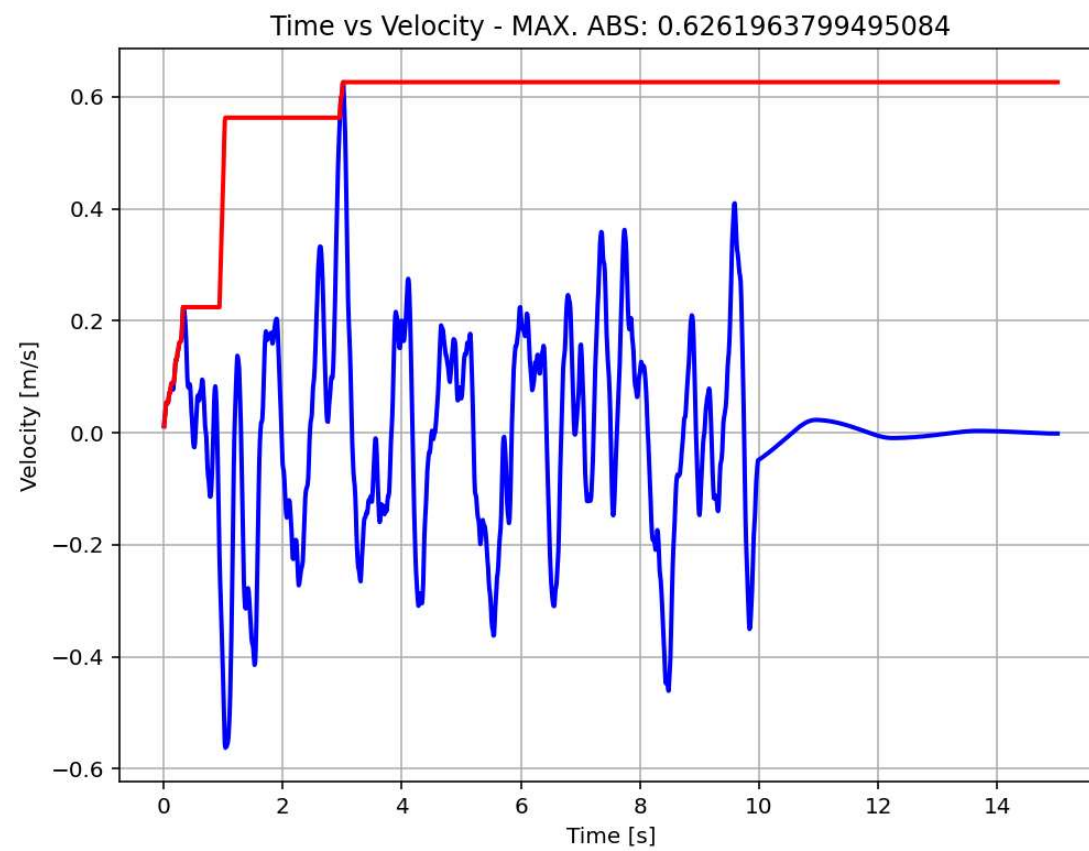
Fragility Curves

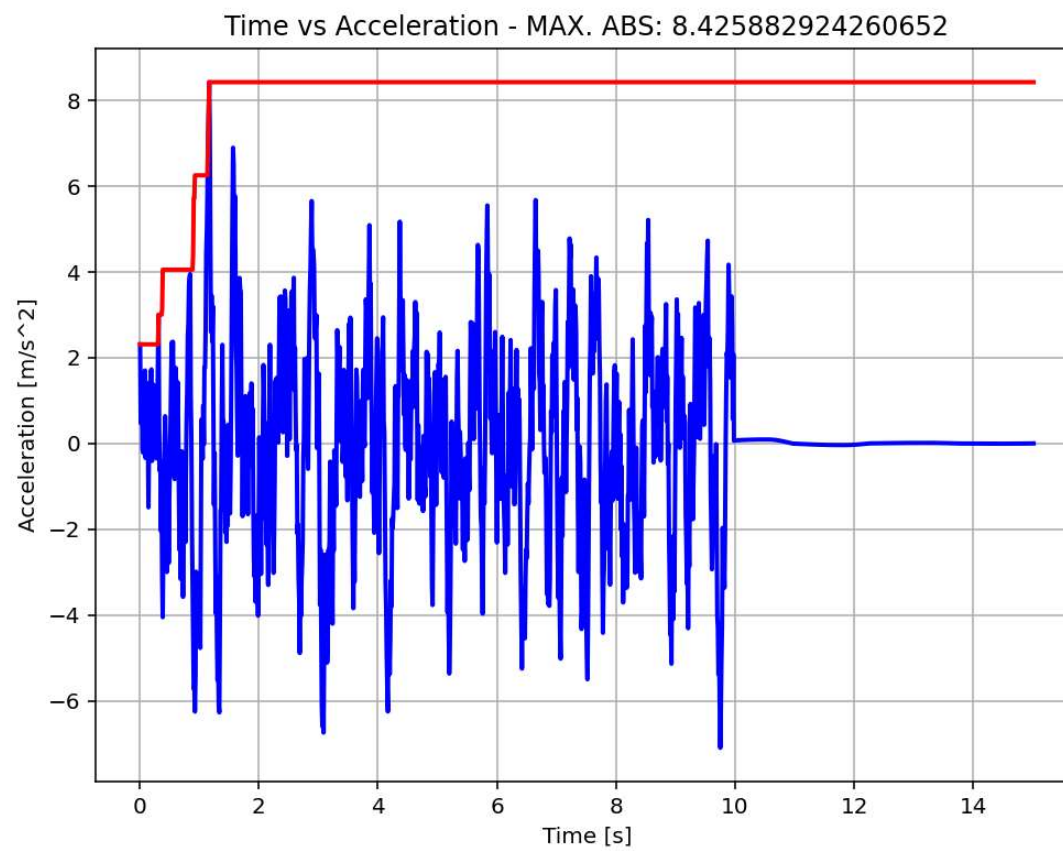


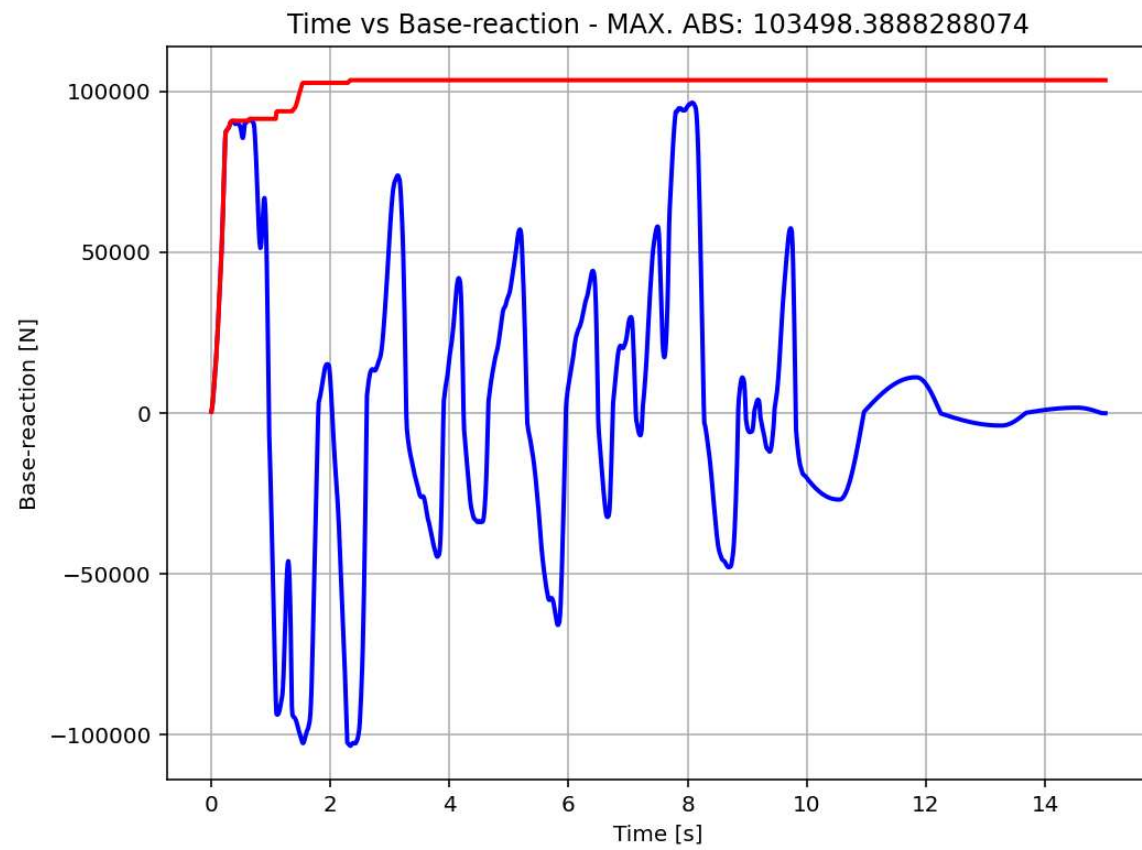
Displacement & Base Reaction Relation From Last Dynamic Analysis

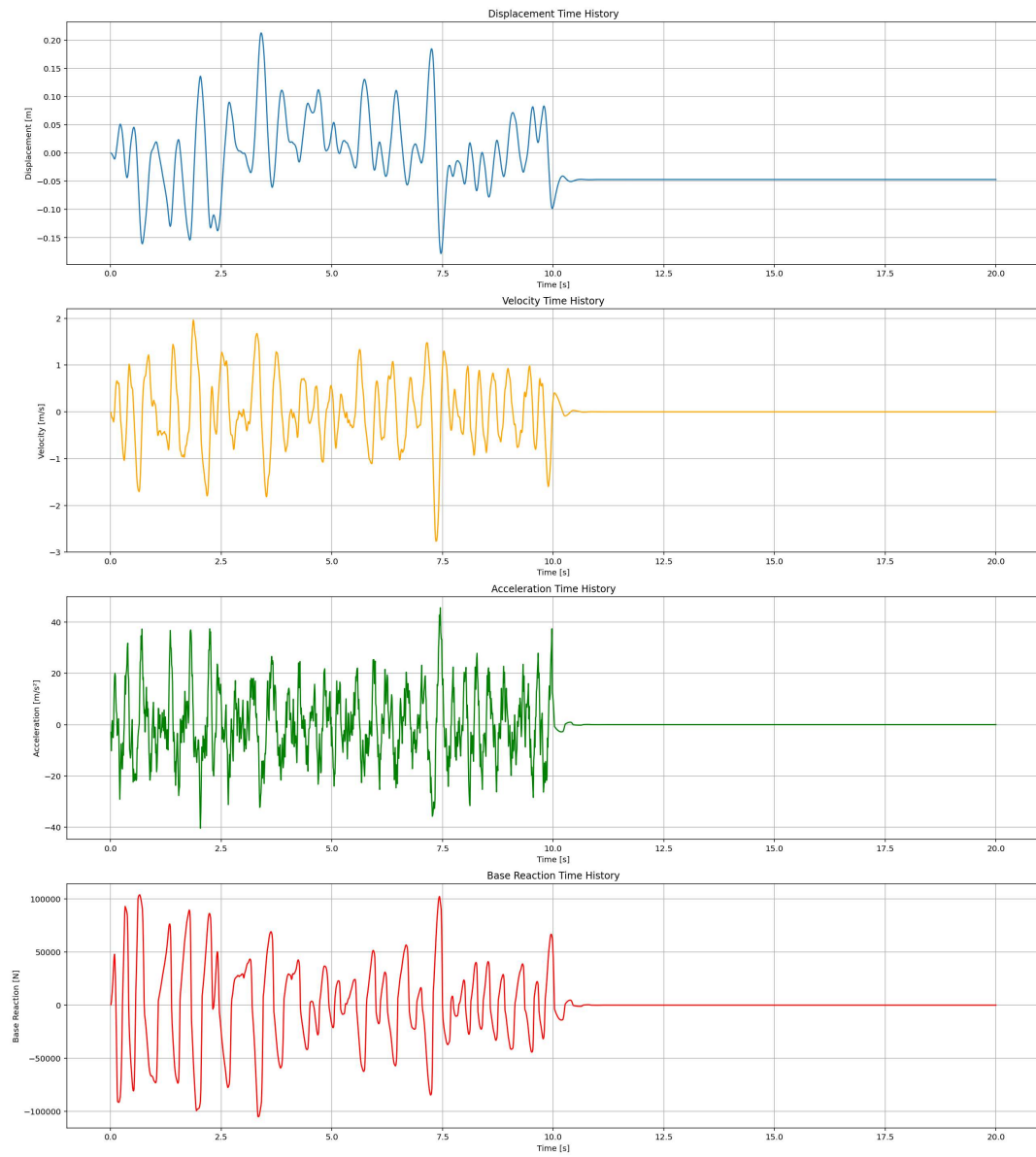












None

